

Xcelr8 Developer Comprehensive Reference (Laravel 12 / PHP 8.3)

1. COMPOSER ESSENTIALS

Command	Usage / Tips
composer install	Install from lock file. Use after cloning/switching.
composer update	Update dependencies & lock file. Use for intentional refreshes.
composer dump-autoload	Rebuild autoloader. Use after adding new classes.
composer require <pkg>	Add a package. Prefer specific versions for team sync.
composer remove <pkg>	Remove a package. Keeps json/lock in sync.
composer validate	Validate composer files before commits.
composer outdated	Show outdated packages for hygiene checks.
composer show	Inspect exact versions of installed packages.
composer diagnose	Run when Composer behaves unexpectedly.
composer exec	Run vendor binaries from project context.

2. ARTISAN (LARAVEL 12)

Command	Usage / Tips
php artisan list	Show all commands. Use if you forget a name.
php artisan help <cmd>	Show command options/flags (e.g. help migrate).
make:model <Name> -mcr	Scaffold Model + Migration + Controller + Resource.
make:controller <Name> --resource	Create a standard CRUD resource controller.
make:request <Name>	Create form request for validation logic.
migrate	Run migrations. Check branch state first.
migrate:fresh --seed	Destructive: Rebuild DB and run all seeders.
migrate:rollback	Undo the last migration operation.
db:seed	Run seeders after fresh migration or data change.
route:list	Inspect all routes for collisions and URIs.
optimize:clear	Best first fix for stale config, routes, or views.
config:cache / clear	Cache config for production / Clear if stale.
route:cache / clear	Cache routes (if safe) / Clear if stale.
view:cache / clear	Cache compiled views / Clear after Blade changes.
queue:work	Process background jobs (keep terminal open).
schedule:work	Run scheduler loop for local testing of jobs.

3. TINKER PATTERNS & CODE GEN

Task	Example / Notes
User::find(1)	Quick inspection of model data.
\$user->orders	Debug Eloquent relationships.
Invoice::overdue()->count()	Validate business logic/scopes.
Bus::dispatch(new Job())	Test background jobs reliably.
config('app.env')	Confirm current environment behavior.
DB::table('u')->count()	Fast database probing.

4. MYSQL EVERYDAY COMMANDS

Command	Use / Tip
SHOW DATABASES;	List available databases.
USE <db_name>;	Switch context. Confirm before writing.
SHOW TABLES;	List tables in current database.
DESCRIBE <table_name>;	Fast column inspection.
SHOW CREATE TABLE <name>;	Full DDL - useful for keys/debugging.
EXPLAIN SELECT...	Inspect query plan and costs.
EXPLAIN ANALYZE <sql>	MySQL 8: Actual execution metrics.
ANALYZE TABLE <name>;	Refresh stats after large data changes.
SELECT / INSERT / UPDATE / DELETE	Standard CRUD. Always use WHERE for delete/update.

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5. GIT COMMANDS & RECOVERY

Command	Action / Habit
git status	Check current state. Run constantly.
git branch	List local branches.
git switch <name>	Switch branches.
git switch -c <name>	Create and switch to new branch.
git pull / git fetch	Fetch & Merge / Download remote refs only.
git add .	Stage all changes. Review with status first.
git commit -m 'msg'	Commit changes. Use short imperative messages.
git push	Send commits to remote. Check branch first.
git log --oneline --graph	View visual history.
git diff / --staged	See unstaged vs staged changes.
git restore <file>	Discard local file edits (Safe rollback).
git restore --staged <file>	Unstage a file but keep edits.
git reset --soft HEAD~1	Undo last commit but keep your changes.
git commit --amend	Modify the last local commit (msg or files).
git log -- <file>	View history of a single file.

6. VS CODE SHORTCUTS & WORKFLOWS

Shortcut	Action
Ctrl + Shift + P	Command Palette - reach any feature.
Ctrl + P	Quick Open - find files by name.
Ctrl + Shift + F	Global Search - find routes/classes/strings.
Ctrl + /	Toggle line comment.
Ctrl + Shift + K	Delete current line.
Alt + Up/Down	Move line up or down.
Ctrl + D / Ctrl + Shift + L	Select next match / Select all matches.
Alt + Shift + F	Format document.
Ctrl + `	Toggle integrated terminal.
Jump to Class	Ctrl + P and type name.
Compare Files	Select in Explorer -> Right Click -> Compare.

7. NAMING & CODING STANDARDS

Artifact	Convention	Example
PHP Classes	PascalCase	PurchaseOrder, InvoiceController
Methods / Variables	camelCase	storeInvoice(), \$isActive
Database Tables	plural_snake_case	purchase_orders, users
Database Columns	snake_case	customer_id, approved_at
Route URIs	lowercase kebab-case	/purchase-orders/{id}
Route Names	dot notation	purchase-orders.show
Blade Files	kebab-case	edit-line-items.blade.php
Models	Singular PascalCase	Customer, InvoiceItem
Controllers	Singular Res + Controller	CustomerController
Requests	Action + Res + Request	StoreCustomerRequest
Jobs / Events	Domain Specific	GenerateInvoicePdfJob

8. ARCHITECTURE & QUALITY GATES

Core Principles:

- Thin Controllers: Coordinate HTTP only. No complex logic. [cite: 2]
- Services/Actions: Logic > 5 lines belongs in dedicated classes. [cite: 2]
- Form Requests: Mandatory for non-trivial validation. [cite: 2]
- Models: Relations, scopes, casts only. [cite: 2]
- N+1 Prevention: Use eager loading for all relationship loops. [cite: 2]

Quality Checklist & Tools:

- Tools: Laravel Pint (Format), Larastan (Analysis), SonarQube (Smells/Gates). [cite: 8]
- Pre-Push: git status clean, No debug dumps (dd/dump), Tests passing. [cite: 2]
- Commits: Short imperative messages (e.g., 'Add customer filter'). [cite: 26]